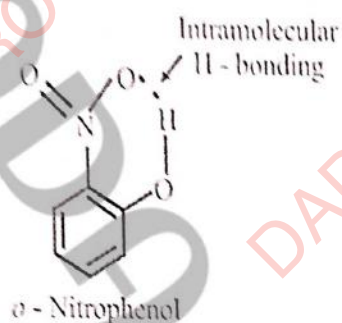


INTRAMOLECULAR HYDROGEN BOND:

This type of hydrogen bonding occurs between the atoms of same molecule



EFFECT OF HYDROGEN BONDING ON PHYSICAL PROPERTIES:

Hydrogen bonding effect on the physical properties of the compound such as;

- ✓ Melting point & boiling point
- ✓ Physical state
- ✓ Solubility

CHARACTERISTICS OF HYDROGEN BOND:

• BOND ENERGIES:

Hydrogen bond is a weaker bond than the covalent bond but stronger than Vander walls forces. As compared to covalent bond the bond energy of hydrogen bond is only one tenth (1/10). It is between 3 to 10 kcal/mole.

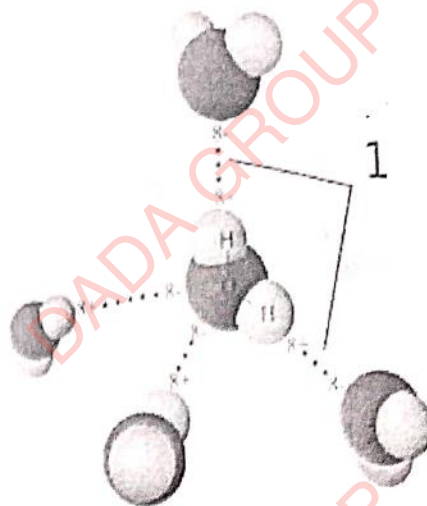
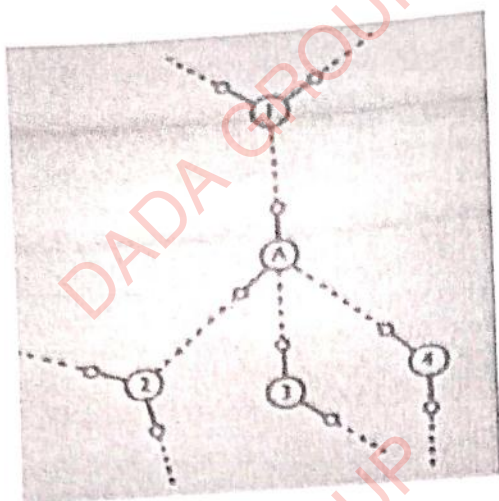
• BOND LENGTH:

It depends upon the nature of molecules. In case of water, the O-H...O distance is $2.76 \text{ \AA}$, In HF, the F...H...F, the distance is $2.55 \text{ \AA}$

ANOMALOUS BEHAVIOUR OF WATER:

When liquids get cooled the molecules come closer to each other and they contract and density become high, when liquid gets heated their density become low till they all become in liquid form. But water shows abnormal behavior i.e. when water heated from 0°C to 4°C it contracts and when cooled from 4°C to 0°C , it expands (contrary to the behavior of other liquids).

This behavior is due to the hydrogen bonding. Crystal structure of ice shows a tetrahedral arrangement of H_2O molecules. Each Oxygen 'A' is surrounded tetrahedrally by four others 1, 2, 3, 4. The hydrogen bonds showed by dotted lines link the pairs of oxygen atoms.



The arrangement of the water molecules in an ice is an open structure and due to larger volume, ice is less dense than liquid water, therefore ice floats on water. When ice melts some of the hydrogen bonds are broken and the molecules pack closely together so that water has a higher density. This breaking down process is not complete until a temperature of 4°C is reached at which water has the maximum density.

HYDROGEN BONDING AND DAILY LIFE:

Hydrogen bond plays important role in the daily life, as our body contains most of organic compounds like (proteins, polymers, etc.). Because of their weak hydrogen bonds, they form and decompose easily at normal temperature as the biological processes in living bodies have to function at normal temperature; this is one of the most vital importance of hydrogen bond in daily life

"HYBRIDIZATION"

DEFINITION:

"The mixing of different atomic orbitals to produce the same number of equivalent orbitals having same shape and energy is known as hybridization and the resulting orbitals are called hybrid orbitals".

EXPLANATION:

- ✓ Only those orbitals take part that have unpaired electron (single electron).
- ✓ Only those orbital take part that has very little difference of energy.
- ✓ Hybridization takes place in the orbitals of a single atom.
- ✓ Two different atoms cannot produce hybrid orbitals by overlapping their orbitals.

HYBRIDIZATION IN CARBON ATOM:

This electronic configuration indicates that carbon is a divalent, but in the majority of compounds it is tetravalent, this contradiction is removed by promoting an electron from $2s$ orbitals to a higher empty p_z orbital.

The carbon is said to be in excited state and behaves as tetravalent. It is an imaginary process, when their unfilled orbitals occupied by the electron of other reacting atoms, a stable configuration is obtained and the new molecule is formed.

TYPES OF HYBRIDIZATION:

The types of hybridization depend upon the number of orbitals which take part in the mixing. Types of hybridization are as follows:

- ✓ $s-p$ hybridization (diagonal)

- ✓ S-p² hybridization (trigonal pyramidal)
- ✓ S-p³ hybridization (tetragonal)

"SP HYBRIDIZATION"

DEFINITION:

"The hybridization of an s- and a p- orbital leads to two hybrid orbitals known as sp-orbitals and this hybridization is known as sp-hybridization."

EXPLANATION:

These sp orbitals have character of s and p in the ratio of 1: 1 and these orbitals are co-linear at an angle of 180° which provides maximum separation and overlap. The hybridized two p-orbitals remain at right angles to the plane of the hybrid orbitals.

SP HYBRIDIZATION IN CARBON:

When carbon combines with other two atoms, then sp hybridization occur. In sp hybridization, one 2s and one 2p orbitals combines to form two hybrid orbitals.

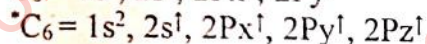
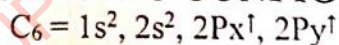
EXAMPLES:

"ETHYNE OR ACETYLENE C₂H₂"

CENTRAL ATOM:

- ✓ Carbon

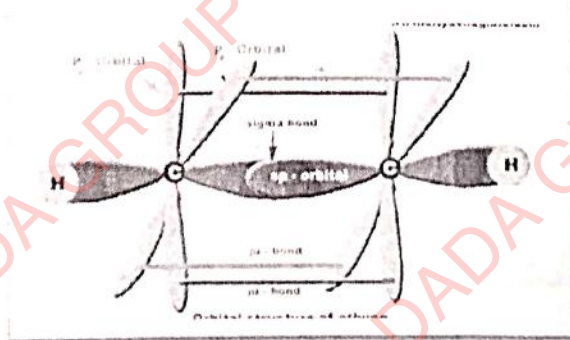
ELECTRONIC CONFIGURATION:



HYBRIDIZATION:

- ✓ In C₂H₂, each carbon atom is S-p hybridized forming two hybrid orbitals.
- ✓ Each carbon atom forms two sigma bonds i.e. one δ_{sp-sp} bond with another Carbon and one δ_{sp-s} bond with hydrogen.
- ✓ In C₂H₂, the remaining Py and Pz orbitals of each carbon are side by side overlapped forming two π -bonds.
- ✓ The shape of Acetylene molecule is linear
- ✓ Bond angle Is 180°.

STRUCTURE:



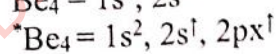
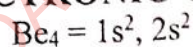
"BERYLLIUM CHLORIDE BECL₂"

CENTRAL ATOM:

- ✓ Beryllium

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH# 3

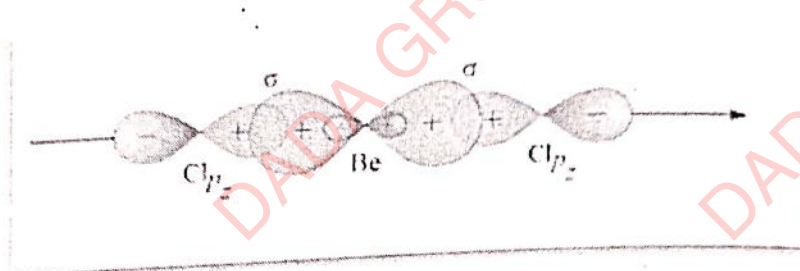
ELECTRONIC CONFIGURATION:



HYBRIDIZATION:

- ✓ In BeCl_2 , beryllium is s-p hybridized forming two hybrid orbitals.
- ✓ In BeCl_2 , beryllium forms two δ_{sp-p} bonds with chlorine.
- ✓ No π -bonds are formed.
- ✓ The shape of molecule of BeCl_2 is linear.
- ✓ Bond angle is 180° .

STRUCTURE:



"S-P² HYBRIDIZATION"

DEFINITION:

"The mixing of one s- and two p- orbitals to produce three sp^2 hybrid orbitals is referred to as sp^2 or trigonal hybridization."

EXPLANATION:

Each sp^2 hybrid orbital has character of s and p orbitals in the ratio of 1:2. They are co-planar and directed towards the corner of an equilateral triangle (trigonal) at angles of 120° to each other. The size of these orbitals is bigger than s- and p- atomic orbitals, therefore it overlaps more effectively and forms stronger bond.

SP² HYBRIDIZATION IN CARBON:

When carbon combines with other three atoms, then sp^2 hybridization occurs. In sp^2 hybridization, one 2s and two 2p orbitals are mixed to form three new hybrid orbitals.

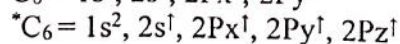
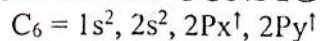
EXAMPLES:

"ETHYLENE C_2H_4 "

CENTRAL ATOM:

- ✓ Carbon

ELECTRONIC CONFIGURATION:



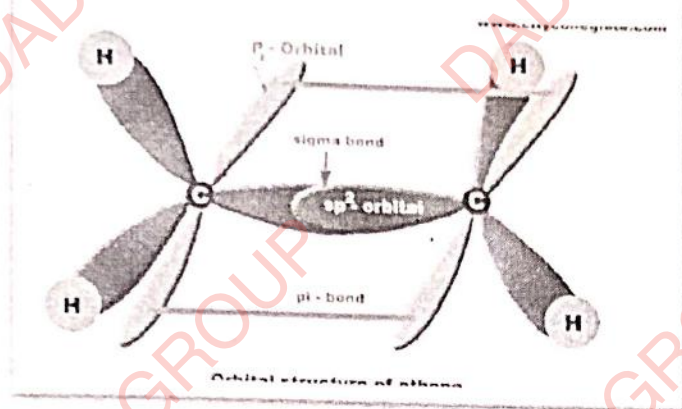
HYBRIDIZATION:

- ✓ In C_2H_4 , each carbon atom is sp^2 hybridized forming three hybrid orbitals.
- ✓ Each carbon atom forms three sigma bonds i.e. one $\delta_{sp^2-sp^2}$ bond with another Carbon and two δ_{sp^2-s} bonds with hydrogen.

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- ✓ In C_2H_4 , the remaining P_z orbitals of each carbon are side by side overlapped forming π -bond.
- ✓ The shape of Acetylene molecule is trigonal planar.
- ✓ Bond angle is 120° .

STRUCTURE:



" BCl_3 "

CENTRAL ATOM:

- ✓ Boron

ELECTRONIC CONFIGURATION:

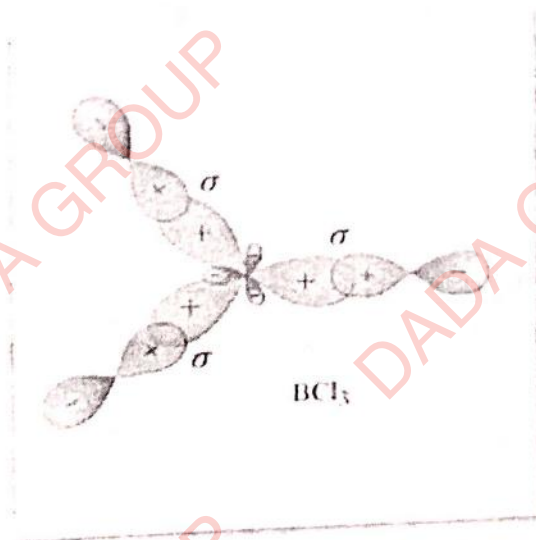
$$B_5 = 1s^2, 2s^2, 2p_x^1$$

$$*B_5 = 1s^2, 2s^1, 2p_x^1, 2p_y^1$$

HYBRIDIZATION:

- ✓ In BCl_3 , boron is sp^2 hybridized forming three hybrid orbitals.
- ✓ In BCl_3 , Boron forms three δ_{s-p^2-p} bonds with chlorine.
- ✓ No π -bonds are formed.
- ✓ The shape of BCl_3 molecule is trigonal planar.
- ✓ Bond angle is 120° .

STRUCTURE:



DADA OFFICAL

"BF₃"

CENTRAL ATOM:

✓ Boron

ELECTRONIC CONFIGURATION:

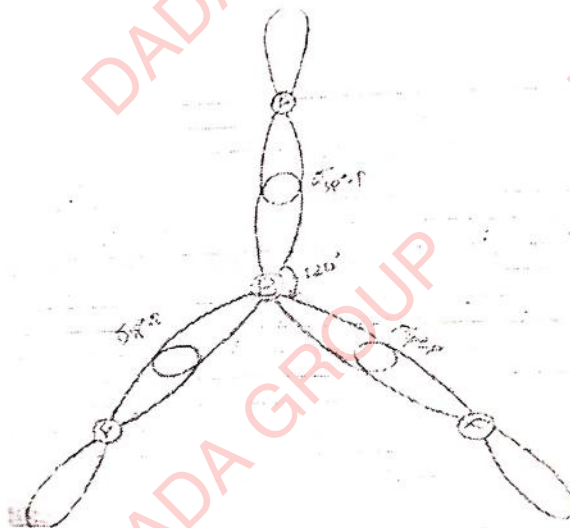
B₅ = 1s², 2s², 2p¹

*B₅ = 1s², 2s¹, 2p¹, 2p¹

HYBRIDIZATION:

- ✓ In BF₃, Boron is S-p² hybridized forming three hybrid orbitals.
- ✓ In BF₃, Boron forms three σ_{sp²-p} bonds with Fluorine.
- ✓ No π-bonds are formed.
- ✓ The shape of BF₃ molecule is trigonal planar.
- ✓ Bond angle is 120°.

STRUCTURE:



"S-P³ HYBRIDIZATION"

DEFINITION:

"Combination of one s- and three p- orbitals to produce four sp³ hybrid orbitals is known as tetrahedral hybridization."

EXPLANATION:

Each sp³ hybrid orbital possess the character of s and p orbitals in the ratio of 1:3. These are directed at the corner of regular tetrahedron with an angle of 109.5°. The size of sp³ orbitals is bigger than s- and p- atomic orbitals, therefore it overlaps more effectively and forms stronger bond.

SP³ HYBRIDIZATION IN CARBON:

When carbon combines with four other atoms then sp³ hybridization occurs. In sp³ hybridization, s- orbital and three p- orbitals are mixed to form four new hybrid orbitals. In sp³ hybridization, the hybrid orbitals arrange themselves in such manner that they acquire tetrahedral structure with a bond angle of 109.28°.

EXAMPLES:

"ETHANE C₂H₆"

CENTRAL ATOM:

✓ Carbon

ELECTRONIC CONFIGURATION:

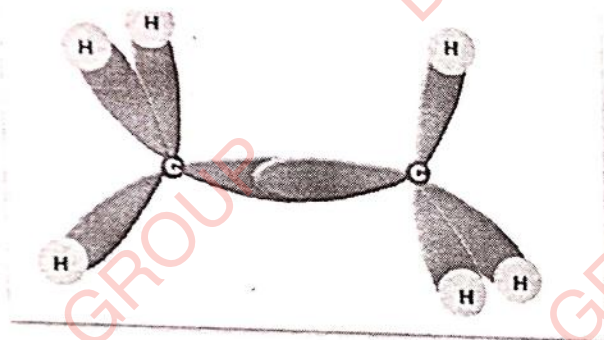
C₆ = 1s², 2s², 2P_x¹, 2P_y¹

*C₆ = 1s², 2s¹, 2P_x¹, 2P_y¹, 2P_z¹

HYBRIDIZATION:

- ✓ In C₂H₆, each carbon atom is S-p³ hybridized forming four hybrid orbitals.
- ✓ Each carbon atom forms four sigma bonds i.e. one δ_{sp^3-s} bond with another Carbon and three δ_{sp^3-s} bonds with hydrogen.
- ✓ In C₂H₆, no π -bonds are formed.
- ✓ The shape of Ethane molecule is tetrahedral.
- ✓ Bond angle is 109.5°.

STRUCTURE:



"METHANE CH₄"

CENTRAL ATOM:

✓ Carbon

ELECTRONIC CONFIGURATION:

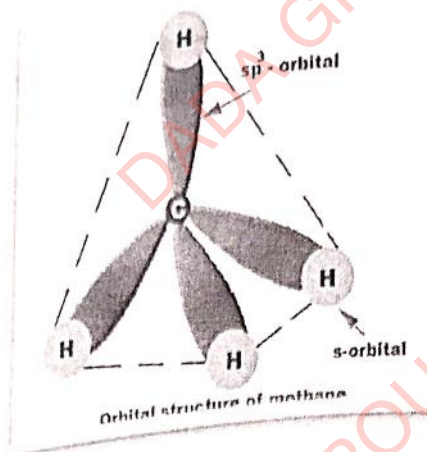
C₆ = 1s², 2s², 2P_x¹, 2P_y¹

*C₆ = 1s², 2s¹, 2P_x¹, 2P_y¹, 2P_z¹

HYBRIDIZATION:

- ✓ In CH₄, Carbon atom is S-p³ hybridized forming four hybrid orbitals.
- ✓ Carbon atom forms four sigma bonds i.e. four δ_{sp^3-s} bonds with four hydrogen atoms.
- ✓ In CH₄, no π -bonds are formed.
- ✓ The shape of Methane molecule is tetrahedral.
- ✓ Bond angle is 109.5°.

STRUCTURE:

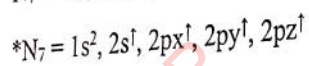
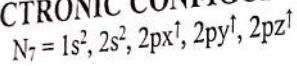


"AMMONIA NH₃"

CENTRAL ATOM:

✓ Nitrogen

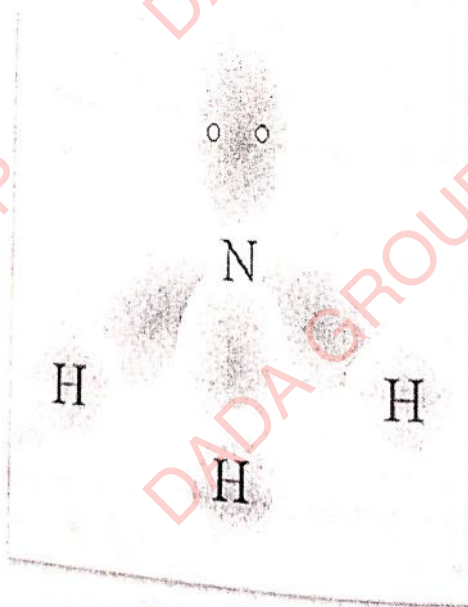
ELECTRONIC CONFIGURATION:



HYBRIDIZATION:

- ✓ In NH₃, Nitrogen is S-p³ hybridized forming four hybrid orbitals.
- ✓ In NH₃, Nitrogen forms three δ_{sp^3-s} bond with three hydrogen atom and has one lone pair.
- ✓ No π -bonds are formed.
- ✓ The shape of ammonia molecule is Pyramidal.
- ✓ Bond angle is 107°.

STRUCTURE:



"PH₃"

CENTRAL ATOM:
✓ Phosphorus

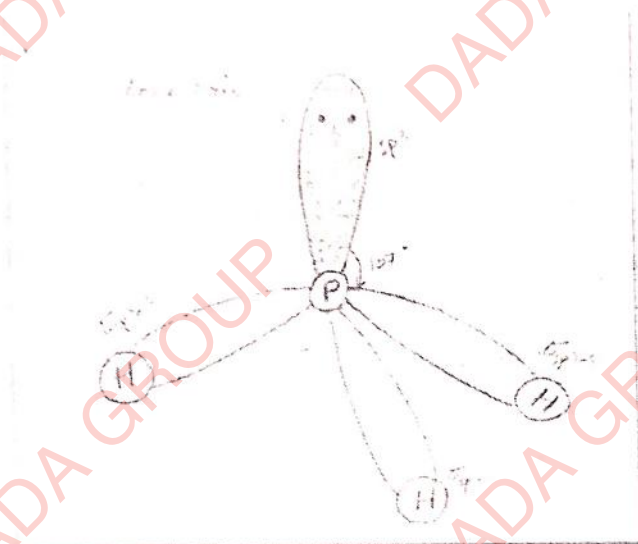
ELECTRONIC CONFIGURATION:
 $P_{15} = 1s^2, 2s^2, 2p^6, 3s^2, 3p_x^1, 3p_y^1, 3p_z^1$

$*P_{15} = 1s^2, 2s^2, 2p^6, 3s^2, 3p_x^1, 3p_y^1, 3p_z^1$

HYBRIDIZATION:

- In PH₃, Phosphorus is S-p³ hybridized forming four hybrid orbitals.
- In PH₃, Phosphorus forms three δ_{sp^3} -s bonds with three hydrogens and has one lone pair.
- No π -bonds are formed.
- The shape of the molecule is pyramidal.
- Bond angle is 107°.

STRUCTURE:



"WATER H₂O"

CENTRAL ATOM:

✓ Oxygen

ELECTRONIC CONFIGURATION:

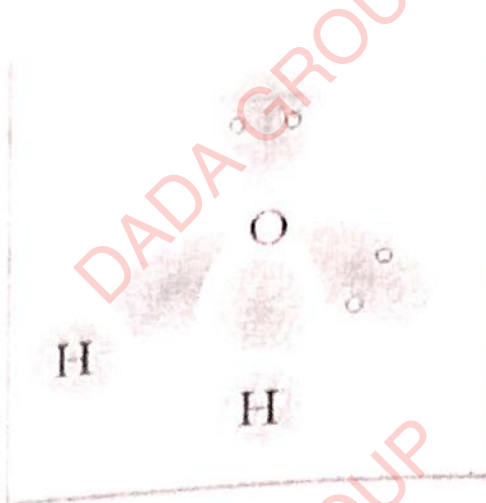
$O_8 = 1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^1$

$*O_8 = 1s^2, 2s^2, 2p_x^1, 2p_y^1, 2p_z^1$

HYBRIDIZATION:

- ✓ In H₂O, oxygen is S-p³ hybridized forming four hybrid orbitals.
- ✓ In H₂O, oxygen forms two δ_{sp^3} -s bond with two hydrogens and has two lone pairs.
- ✓ No π -bonds are formed.
- ✓ The shape of water molecule is bent.
- ✓ Bond angle is 104°.

STRUCTURE:



"H₂S"

CENTRAL ATOM:

✓ Sulphur

ELECTRONIC CONFIGURATION:

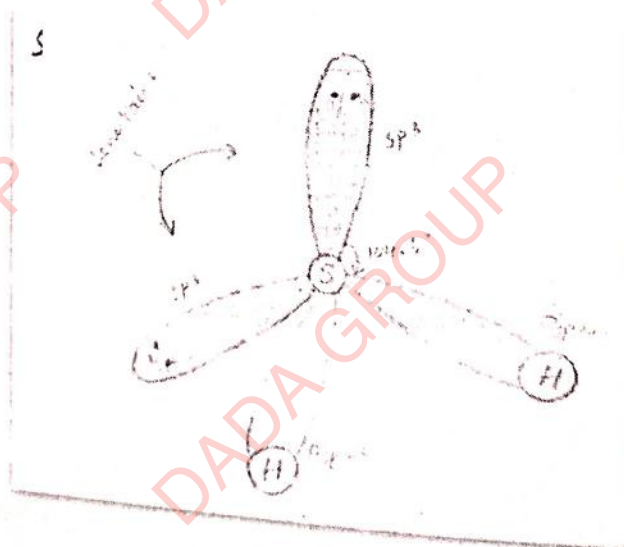
$$S_{16} = 1s^2, 2s^2, 2p^6, 3s^2, 3p_x^{\uparrow}, 3p_y^{\uparrow}, 3p_z^{\uparrow}$$

$$*S_{16} = 1s^2, 2s^2, 2p^6, 3s^2, 3p_x^{\uparrow}, 3p_y^{\uparrow}, 3p_z^{\uparrow}$$

HYBRIDIZATION:

- In H₂S, Sulphur is S-p³ hybridized forming four hybrid orbitals.
- In H₂S, Sulphur forms two δ_{sp³-s} bonds with two hydrogens and has two lone pairs.
- No π-bonds are formed.
- The shape of molecule is bent.
- Bond angle is 104°.

STRUCTURE:



"F₂"

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH# 3

CENTRAL ATOM:
✓ Fluorine

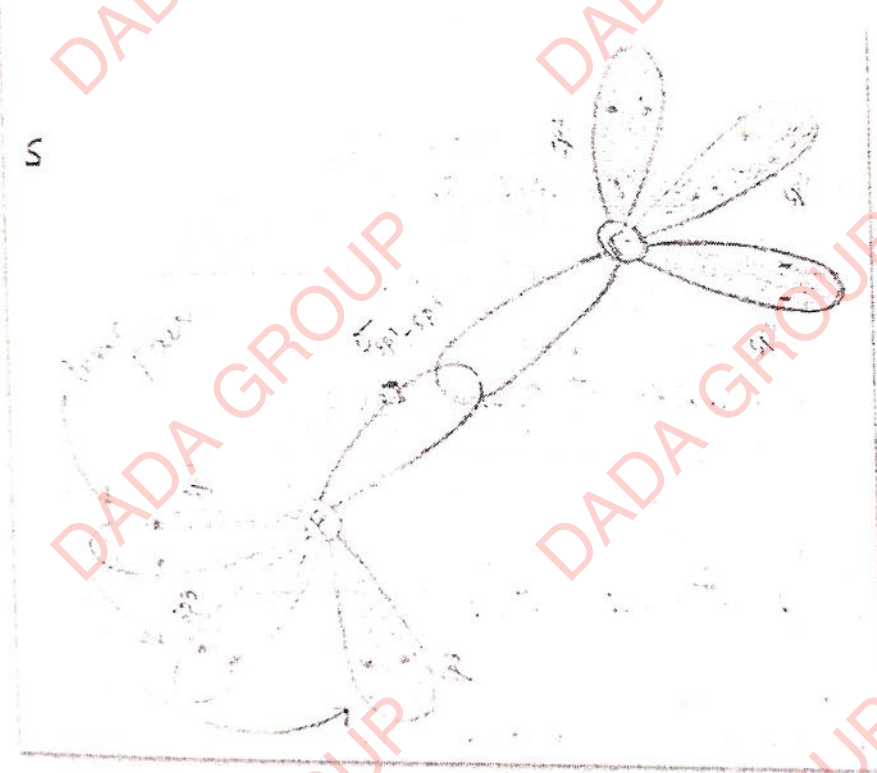
ELECTRONIC CONFIGURATION:
 $F_9 = 1s^2, 2s^2, 2p_x^{\uparrow\downarrow}, 2p_y^{\uparrow\downarrow}, 2p_z^{\uparrow}$

$$*F_9 = 1s^2, 2s^2, 2p_x^{\uparrow\downarrow}, 2p_y^{\uparrow\downarrow}, 2p_z^{\uparrow}$$

HYBRIDIZATION:

- ✓ In F_2 molecule, each fluorine is $s-p^3$ hybridized forming four hybrid orbitals.
- ✓ In F_2 molecule, each fluorine forms only one $\delta_{sp^3-sp^3}$ bond with another fluorine and has three lone pairs.
- ✓ No π -bonds are formed.
- ✓ The shape of the molecule is linear.
- ✓ Bond angle is 180° .

STRUCTURE:



"STRUCTURES BASED ON V.S.E.P.R.T"

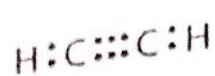
"ETHYNE OR ACETYLENE C_2H_2 " ✓

CENTRAL ATOM:

✓ Carbon

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH 1

LEWIS STRUCTURE:



or



A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 2
- ✓ Bond pair: 2
- ✓ Lone pairs: nil
- ✓ π -bonds: 2
- ✓ Shape: Linear
- ✓ Bond angle: 180°

STRUCTURE:



"ETHENE C_2H_4 " ✓

CENTRAL ATOM:

- ✓ Carbon

LEWIS STRUCTURE:

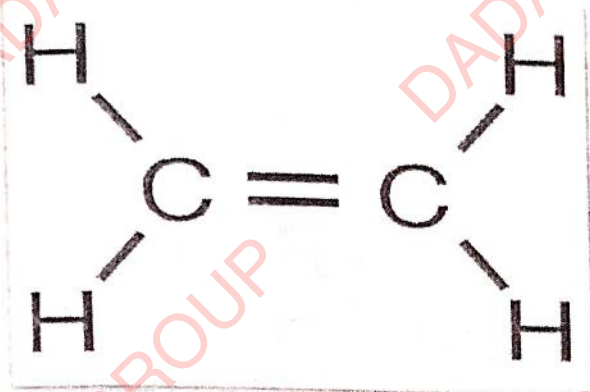


A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 3
- ✓ Bond pair: 3
- ✓ Lone pairs: nil
- ✓ π -bonds: 1
- ✓ Shape: Trigonal planar
- ✓ Bond angle: 120°

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH# 3

STRUCTURE:

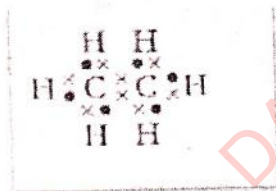


"ETHANE C_2H_6 "

CENTRAL ATOM:

✓ Carbon

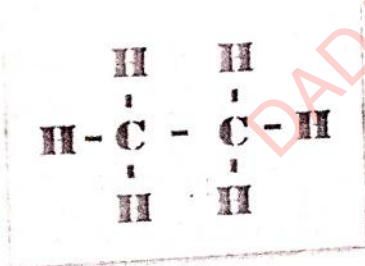
LEWIS STRUCTURE:



A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 4
- ✓ Lone pairs: nil
- ✓ Shape: Tetrahedral
- ✓ Bond angle: 109.28°

STRUCTURE:

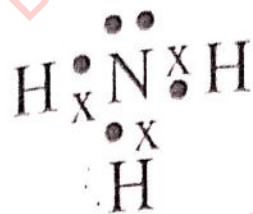


"AMMONIA NH₃"

CENTRAL ATOM:

✓ Nitrogen

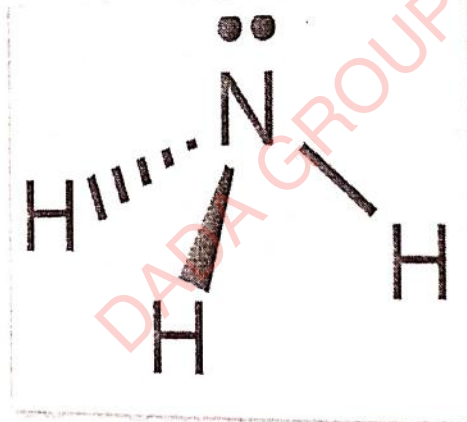
LEWIS STRUCTURE:



A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 3
- ✓ Lone pairs: 1
- ✓ Shape: Pyramidal
- ✓ Bond angle: 107°

STRUCTURE:

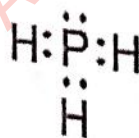


"PH₃"

CENTRAL ATOM:

✓ Phosphorus

LEWIS STRUCTURE:



Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH# 3

A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 3
- ✓ Lone pairs: 1
- ✓ Shape: Pyramidal
- ✓ Bond angle: 107°

STRUCTURE:

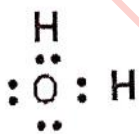


"WATER H₂O"

CENTRAL ATOM:

- ✓ Oxygen

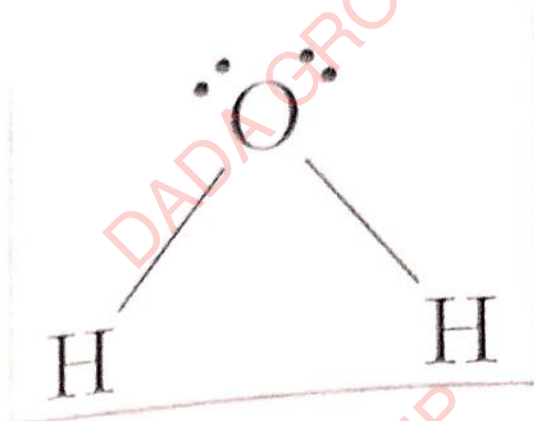
LEWIS STRUCTURE:



A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 2
- ✓ Lone pairs: 2
- ✓ Shape: Bent
- ✓ Bond angle: 104.5°

STRUCTURE:

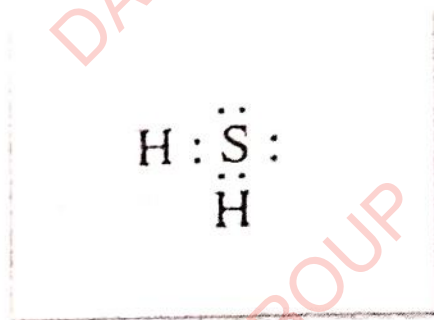


"H₂S"

CENTRAL ATOM:

✓ Sulphur

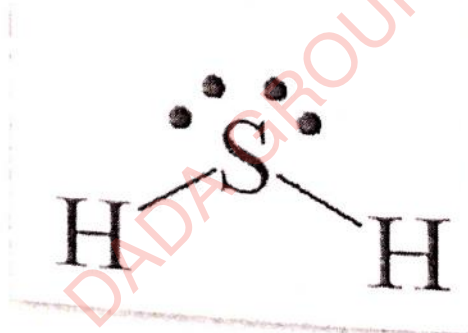
LEWIS STRUCTURE:



A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 2
- ✓ Lone pairs: 2
- ✓ Shape: Bent
- ✓ Bond angle: 104.5°

STRUCTURE:



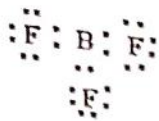
"BF₃"

CENTRAL ATOM:

✓ Boron

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CII# 3

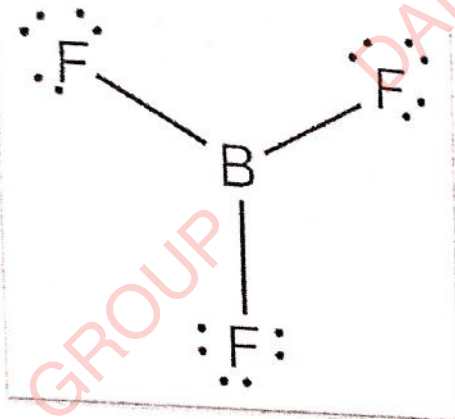
LEWIS STRUCTURE:



AC TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 3
- ✓ Bond pair: 3
- ✓ Lone pairs: Nil
- ✓ Shape: Trigonal planar
- ✓ Bond angle: 120°

STRUCTURE:

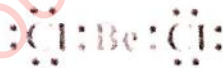


"BeCl₂"

CENTRAL ATOM:

- ✓ Beryllium

LEWIS STRUCTURE:



AC TO V.S.E.P.R.T:

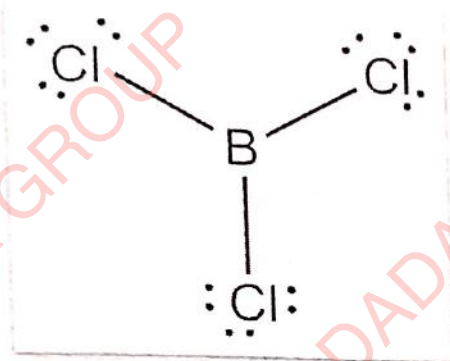
- ✓ Active sets of electrons: 2
- ✓ Bond pair: 2
- ✓ Lone pairs: Nil
- ✓ Shape: Linear
- ✓ Bond angle: 180°

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH# 3

A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 3
- ✓ Bond pair: 3
- ✓ Lone pairs: Nil
- ✓ Shape: Trigonal planar
- ✓ Bond angle: 120°

STRUCTURE:

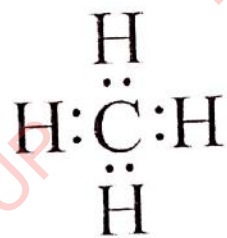


"METHANE CH₄"

CENTRAL ATOM:

- ✓ Carbon

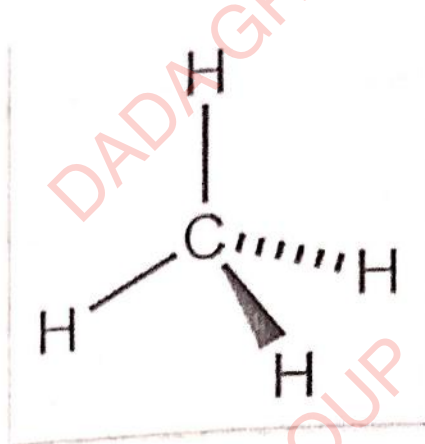
LEWIS STRUCTURE:



A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 4
- ✓ Lone pairs: nil
- ✓ Shape: Tetrahedral
- ✓ Bond angle: 109.28°

STRUCTURE:

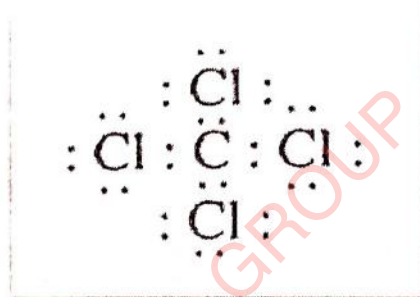


"CCl₄"

CENTRAL ATOM:

✓ Carbon

LEWIS STRUCTURE:

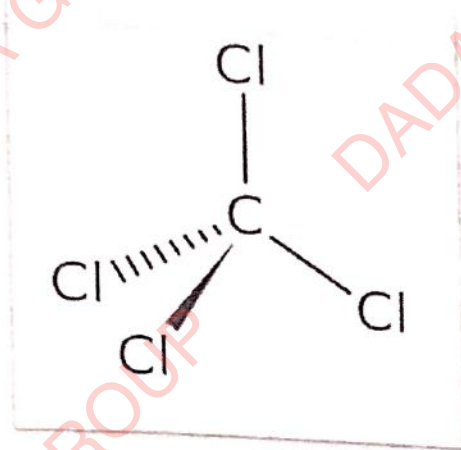


A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 4
- ✓ Lone pairs: nil
- ✓ Shape: Tetrahedral
- ✓ Bond angle: 109.28°

DADA OFFICAL

STRUCTURE:

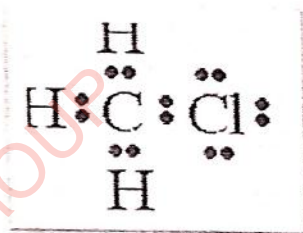


" CH_3Cl "

CENTRAL ATOM:

✓ Carbon

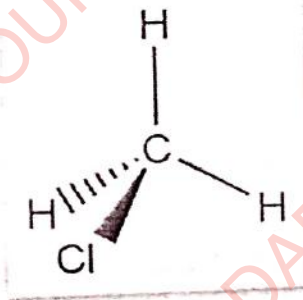
LEWIS STRUCTURE:



AC TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 4
- ✓ Lone pairs: nil
- ✓ Shape: Tetrahedral
- ✓ Bond angle: 109.28°

STRUCTURE:



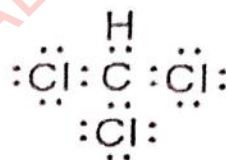
" CHCl_3 "

CENTRAL ATOM:

✓ Carbon

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH

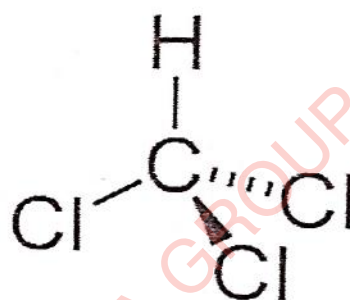
LEWIS STRUCTURE:



A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 4
- ✓ Lone pairs: nil
- ✓ Shape: Tetrahedral
- ✓ Bond angle: 109.28°

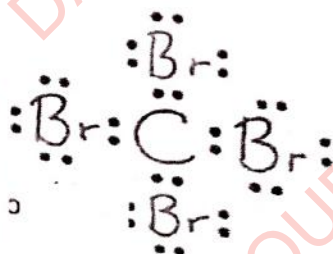
STRUCTURE:



CENTRAL ATOM:
✓ Carbon

LEWIS STRUCTURE:

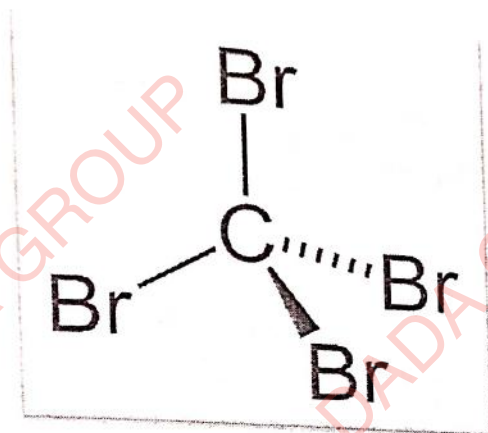
" CBr_4 "



A/C TO V.S.E.P.R.T:

- ✓ Active sets of electrons: 4
- ✓ Bond pair: 4
- ✓ Lone pairs: nil
- ✓ Shape: Tetrahedral
- ✓ Bond angle: 109.28°

STRUCTURE:



"EFFECT OF BONDING ON PHYSICAL AND CHEMICAL PROPERTIES"

The nature and strength of bond affect on various physical and chemical properties of a substance such as solubility, melting point, boiling point, rate of reaction etc.

SOLUBILITY OF POLAR AND NON POLAR COVALENT COMPOUNDS

It is well established that certain substances dissolve in liquid solvent while others are not. The criteria for deciding of solubility is to know whether they are polar or non polar.

• SOLUBILITY OF POLAR SUBSTANCE IN POLAR SOLVENT

Polar covalent molecule is as a whole neutral but it possesses partially positive and negative charge on its two ends. Thus when it is added into a polar solvent a dipole-dipole interaction is developed among, solute-solvent and particles which makes it soluble.

• SOLUBILITY OF NON POLAR SUBSTANCE IN POLAR SOLVENT

Non polar covalent molecules do not dissolve in polar solvent. For example when benzene put into water, it does not dissolve and form two layers. The reason is that benzene is non polar whereas water is polar in nature. Thus the positive and negative poles of water behave hydrophobically with benzene and not allow to dissolving it..

• SOLUBILITY OF NON POLAR SUBSTANCE IN NON POLAR SOLVENT

Non polar covalent molecules dissolve in non polar solvent on the basic principle that like dissolves like due to London forces. For example oil is dissolved in hexane because both are non polar. Since interaction among the particles of both oil and hexane are weak and comparable thus on dissolution, the attractive forces of solvent are enough for overcoming the attractive forces among oil molecules hence it dissolves.

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES"

REACTIONS OF POLAR AND NON POLAR COVALENT COMPOUNDS

Chemical reactions take place by the breaking of existing bonds and the formation of some new bonds. Breaking of bond requires energy. The stronger the bond in the reactant molecule the higher the energy needed for their breaking. Since the atoms in the polar covalent molecule are held more firmly, the bond dissociation energy is required to proceed the reaction. The feasibility of a chemical reaction is determined by having the information of bond energy required to break the total number of reactant molecules and the bond energy release in bond formation in total number of product molecules.

"DIFFERENCES"

SIGMA BOND AND PI BOND:

<u>SIGMA BOND</u>	<u>PI BOND</u>
1. The bond, which is formed by the linear overlapping of atomic orbitals, is called sigma bond. 2. This bond is formed by s-s, s-p and p-p orbitals. 3. The electron cloud density is maximum on the bond axis joining the two nuclei. 4. It is a strong bond. 5. It is less reactive. 6. Its bond energy is higher.	1. The bond, which is formed by the parallel overlapping of atomic orbitals, is called pi bond. 2. This bond is formed by the two parallel p-orbitals. 3. The electron cloud density lies above and below the bond axis. 4. It is a weak bond 5. It is less reactive. 6. Its bond energy is lower.

ATOMIC ORBITAL AND MOLECULAR ORBITAL:

<u>ATOMIC ORBITAL</u>	<u>MOLECULAR ORBITAL</u>
1. It surrounds only one nucleus. 2. It does not indicate the shape of a molecule. 3. Atomic orbitals can be hybridized.	1. It surrounds two or more nuclei. 2. It indicates the shape of a molecule. 3. It cannot be hybridized.

POLAR AND NON POLAR COVALENT BOND:

<u>POLAR COVALENT BOND</u>	<u>NON POLAR COVALENT BOND</u>
1. It is formed between the two atoms, having different electronegative values. 2. Electron pair is shared unequally between the two bonded atoms. 3. Each of the two bonded atoms acquires partial charges.	1. It is formed between the two atoms, having same electronegative values. 2. Electron pair is shared equally between the two bonded atoms. 3. No partial charges are appeared on any atom.

BONDING AND ANTI BONDING MOLECULAR ORBITALS:

BONDING MOLECULAR ORBITAL

1. Its energy is less than that of the parent individual atomic orbitals.
2. It is more stable than that of the parent individual atomic orbitals.
3. Probability of finding the electrons is maximum in it.

ANTIBONDING MOLECULAR ORBITAL

1. Its energy is greater than that of the parent individual atomic orbitals.
2. It is less stable than that of the parent individual atomic orbitals.
3. Probability of finding the electrons is minimum in it.

"OBJECTIVES"

1. A bond formed due to transference of electron is called _____.
2. A bond formed due to mutual sharing of electrons is called _____.
3. Sigma bond is _____ than pi bond.
4. The shape of methane molecule is _____.
5. One s and three p orbitals overlap to produce four _____ hybrid orbitals.
6. Ethene, C_2H_4 is an example of _____ hybridization.
7. Water molecule has _____ structure.
8. Water molecules are inter-linked with one another due to _____.
9. Polarity of the molecule is due to the difference of _____ between the two bonded atoms.
10. A chemical bond formed between two different atoms by mutual sharing of electron is termed as _____.
11. A chemical bond formed between two similar atoms by mutual sharing of electrons is known as _____.
12. The difference between the Electronegativity values of the two atoms forming covalent bond must be _____ than 1.7.
13. Covalent bond in Cl_2 molecule is _____.
14. _____ bond is formed when p-orbitals of the two atoms with their axes parallel to each other overlap with each other.
15. Melting and boiling point of ionic compounds are usually _____ than that of covalent compounds.
16. Non polar compounds are usually _____ in non-polar solvent.
17. The nitrogen in NH_3 is _____ hybridized.
18. H-H bond is _____ than H-Cl bond.
19. Since dipole moment of CS_2 is zero, it is a _____ molecule.
20. A bond formed due to the electrostatic forces of attraction between the oppositely charged ions is called _____ bond.
21. The ionic bond is formed between the atoms with low ionization potential and high _____.
22. A bond formed by the sharing of an electron pair contributed by one atom only is called a _____ bond.
23. A co-ordinate covalent bond is also known as _____ bond.
24. Polar covalent bond is _____ than a non-polar covalent bond.
25. H-F bond is _____ than H-Br bond.
26. The SI unit of dipole moment is _____.
27. Commonly used unit of dipole moment is _____.
28. Dipole moment of non-polar compound is _____.
29. H-H bond is _____ than H-Cl bond.
30. CO_2 is non-polar because of its _____ structure.

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH 1

31. Ionic compounds are generally _____ in nature.
32. A covalent bond is represented by a _____.
33. A co-ordinate covalent bond is represented by an _____.
34. The covalent bond between H-F is called _____ covalent bond.
35. Water molecule has dipole moment because of its _____ structure.
36. The sum of total number of electron pairs (bonding and lone pairs) is called _____ number.
37. A pair of electrons residing on the central atom and which is not used in bonding is called a _____.
38. _____ bond energy than a single bond.
39. An orbital which surrounds a single nucleus is called _____ orbital.
40. An orbital which surrounds two or more atomic nuclei is called _____ orbital.
41. A molecular orbital, which is of lower energy than the atomic orbitals from which it is derived is known as _____ orbital.
42. A molecular orbital, which has higher energy than the atomic orbitals from which it is derived is known as _____ orbital.
43. Bond angle in sp^3 hybridization is of _____.
44. Bond angle in sp^2 hybridization is of _____.
45. Bond angle in sp hybridization is of _____.
46. sp^3 hybridization is also known as _____.
47. sp^2 hybridization is also known as _____.
48. sp hybridization is also known as _____.

"MULTIPLE CHOICE QUESTIONS"

1. The energy required to break a chemical bond to form neutral atoms is called _____.
(Ionization Potential, Electron Affinity, Bond Energy)
2. The chemical bond present in H-Cl is _____.
(Non-Polar, Polar Covalent, Electrovalent)
3. A polar covalent bond is formed between two atoms when the difference between their E.N. value is _____.
(Equal to 1.7, less than 1.7, More than 1.7)
4. The most polar covalent bond out of the following is _____.
(H-Cl, H-F, H-I)
5. _____ bond is one in which an electron has been completely transferred from one atom to another.
(Ionic, Covalent, co-ordinate)
6. _____ bond is one in which an electron pair is shared equally between the two atoms.
(Ionic, Covalent, Co-ordinate)
7. Bond angle in the molecule of CH_4 is of _____.
(120° , 109.5° , 180°)
8. A molecule of CO_2 has _____ structure.
(Linear, trigonal, tetragonal)
9. The sigma bond is _____ than pi bond.
(Weaker, Stronger, Unstable)
10. The sp^3 orbitals are _____ in shape.
(Tetrahedral, Trigonal, Diagonal)
11. The shape of CH_4 molecule is _____.
(Tetrahedral, Trigonal, Diagonal)
12. The bond in Cl_2 is _____.
(Non-polar, Polar, Electrovalent)

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH# 3

1. Water is _____ molecule.
(None polar, Polar, Electrovalent)
2. Covalent bonds in which electron pair are shared equally between the two atoms is called _____ covalent bond.
(Non-polar, Polar, Co-ordinate)
3. Each carbon atom in CH_4 is _____ hybridized.
(Sp^3 , Sp^2 , Sp)
4. Each carbon atom in C_2H_4 is _____ hybridized.
(Sp^3 , Sp^2 , Sp)
5. Each carbon atom in C_2H_2 is _____ hybridized.
(Sp^3 , Sp^2 , Sp)
6. Oxygen atom in H_2O has _____ unshared electron pair.
(One, two, three)
7. Nitrogen atom in NH_3 has _____ unshared electron pair.
(One, two, three)
8. The cloud of charge that surrounds two or more nuclei is called _____ orbital.
(Atomic, Molecular, Hybrid)
9. A substance, which is highly attracted by a magnetic field, is called _____.
(Electromagnetic, Paramagnetic, Diamagnetic)
10. HF exists in liquid due to _____.
(Vander Waal Forces, Hydrogen bond, covalent Bond)
11. Best hydrogen bonding is found in _____.
(HF , HCl , HI)
12. Shape of CCl_4 molecule is _____.
(Tetrahedral, Trigonal, Diagonal)
13. _____ bond is formed due to linear overlap.
(Sigma bond, Pi bond, Hydrogen bond)
14. _____ is defined as the quantity of energy required to break one mole of covalent in gaseous state. (Bond energy, Ionization energy, Energy of Activation)
15. Repulsive force between electron pair in a molecule is maximum when it has an angle of _____.
(120° , 109.5° , 180°)
16. The sum of total number of electrons pairs (bonding and lone pairs) is called _____.
(Atomic Number, Avogadro's number, Steric Number)
17. Shape of _____ molecule is tetrahedral.
(BaCl_2 , BF_3 , NH_3)

THEORY QUESTIONS

Chemistry I "COVALENT BONDING AND SHAPE OF MOLECULES" CH.

- Explain the steps which involved in the formation of sodium chloride on the basis of energy changes.
- Differentiate between atomic orbital and molecular orbital.
- Draw cross and dot structure of the C_2H_4 and CH_3Cl .
- Draw dot and cross structure of CHCl_3 .
- Define ionic bond.
- Write down the characteristics of ionic compounds.
- Define polar and non-polar covalent bond.
- Define Single, Double and Triple covalent bond.
- Define the hydrogen bond.
- What is a coordinate covalent bond? Explain the formation of dative bond in the following:
a. AlCl_3 and Cl^- b. NH_3 and H^+ POCl_3
- Discuss the ionic character of covalent bond.
- Distinguish between the Sigma and Pi bond.
- Draw the structure of the following on the basis of electron pair repulsion theory.
 NH_3 H_2S BeCl_2 CCl_4 CO_3^{2-} C_2H_4
 CS_2
- Define valence band theory?
- State Molecular Orbital Theory (M.O.T.).
- What is Dipole Moment?
- Explain sp^3 hybridization in carbon in details.
- Define hybridization. Write its types and explain any two with examples.
- Explain the shape of ethene on the basis of Hybridization.
- What rule and principle are violated in the following electronic configuration?
 $1s^2, 2s^3, 1s^2, 2p^2$